Supporting Information for : A matter of size: what underpinned the emergence and demise of the largest rodents that ever lived?

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This file contains:

- Supplementary text
- Supplementary figures
- Supplementary tables

Other Supplementary Materials may be found at `<placeholder>`.
Codes are available at: https://github.com/Buffan3369/Chinchilloids.git

Supplementary text

Some details regarding decisions we took during our occurrence data cleaning

Age ranges of the Miocene occurrences from the Río Santa Cruz (Santacrucian SALMA) were sharpened based on their assigned locality (Arnal et al. 2019). Hence, occurrence from Barrancas Blancas were respectively assigned the 16.3–17.21 and 15.3–16.47 Ma intervals.

Given that *Lagostomus (Lagostomopsis)* is a subgenus (Rasia and Candela 2012), we set the `genus` column of each *Lagostomus (Lagostomopsis)* occurrence to *Lagostomus*, and switched their `genus_level_status` from `extinct` to `extant`.

All specimens referred to as *Lagostomus minimus*, *L. cavifrons*, *L. heterogenidens*, *L. debilis* and *L. egenus* were synonymised with *Lagostomus maximus*, following (Ubilla and Rinderknecht 2016; Rasia 2024).

The single occurrence assigned to *Lagostomus striatus* was removed as considered *nomen nodum* by Rasia (2016).

Uncertain age of the Ituzáingo formation was set to Huayquerian SALMA (Late Miocene), *ca.* 5–8 Ma.

Age range of occurrences from Bucher et al. (2021) were set to the one of the original publication: 14.86–12.75 Ma when coming from the "Cruces Infinitos" and "Los Yeguarizos localities", 12.75–12 Ma when coming from the "La Gloria" and "La Hoyada" localities.

Based on Brandoni et al. (2016), the age range of the occurrences from the Pinturas fauna (*e.g.*, (Kramarz 2002; 2004; 2006)), Burdigalian, were set between 20 and 16.8 Ma.

According to (Esteban et al. 2014; Esteban et al. 2019), the age range of the occurrences belonging to the Andalhualá formation was set to Huayquerian–Chapadmalalan (*c.* 7.14–3.66 Ma).

Perimys perpinguis and *Perimys impactus* were lumped in *Perimys onostus* according to McGrath et al. (2022).

Age range of the occurrence belonging to the Sierra Baguales (Santa Cruz formation, Magallanes, Chile) were set to 19–17.8 Ma according to the isotopic constraints provided in Bostelmann et al. (2013).

Age range of the occurrences reported from the Cerro Boleadoras Formation was set to 16.5–15.1 Ma, according to Vizcaino et al. (2022).

Added two occurrences of *Saremmys ligcura* from the two localities of Bryn Gwyn and Gran Barranca (Chubut, Argentina), according to Busker et al. (2019).

We acknowledge the proposition of synonimisation between *Telicomys gigantissimus* and *Telicomys giganteus* made by Rasia et al. (2025). However, in the light of phylogenetic analyses that are currently in progress, we decided not to account for it, therefore treating *T. gigantissimus* and *T. giganteus* as two distinct taxa.

We incorporated four occurrences − corresponding to four different taxa − from the AMD-45 locality (18.9–17.1 Ma) published by Arnal et al. (2026): *Maquiamys praecursor*, *Manumys materdei*, *Marianamys altadens* and *Peruchinchilla prima*.

We followed the latest taxonomic revisions proposed by Rasia (2026) regarding the hyperdiverse dinomyid fauna from the Ituzaingó Formation (Late Miocene).

Additional details are provided in the `Remark` column of *Dataset S1*.

Supplementary Figures

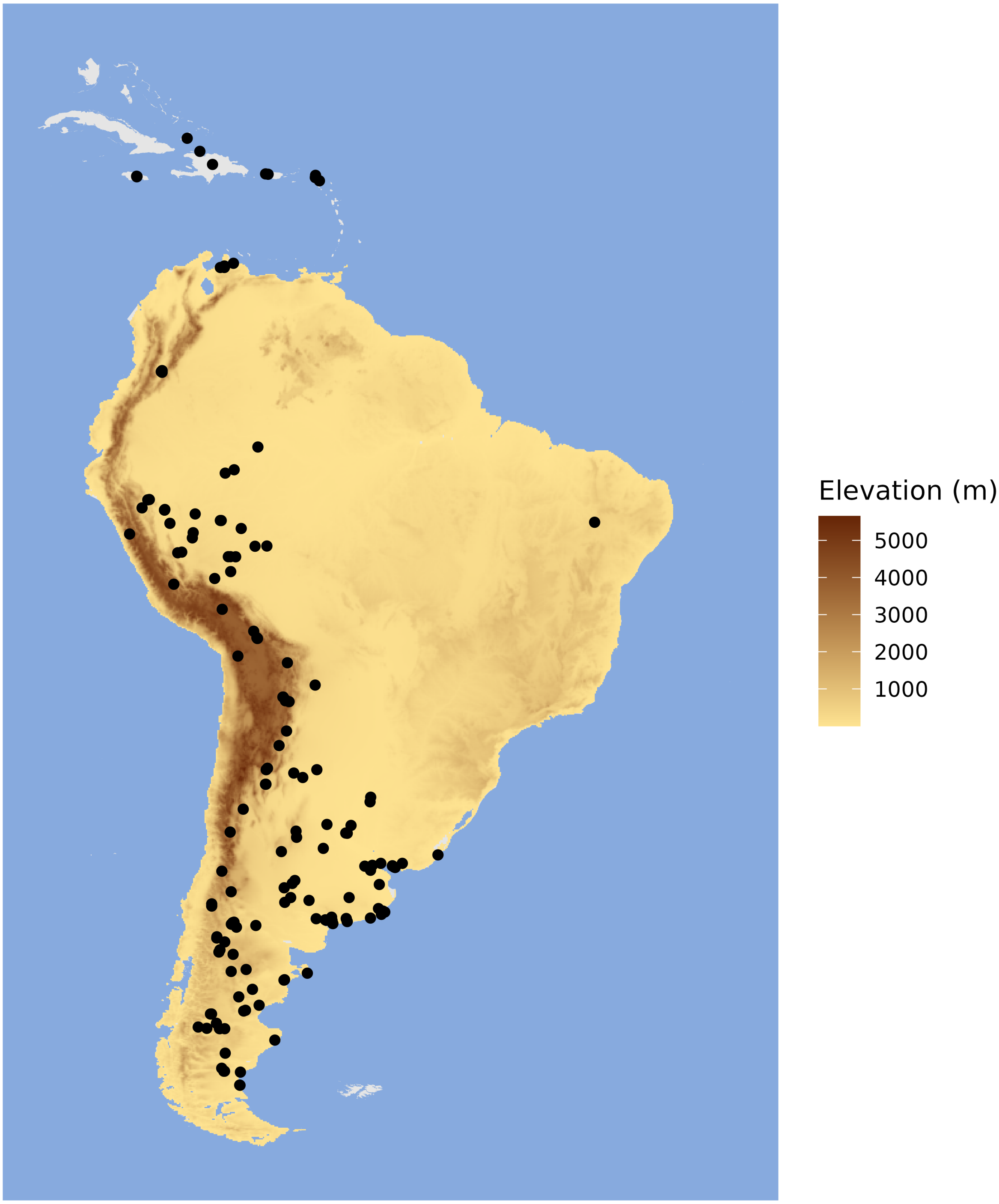

Figure S 1. Location of the chinchilloid bearing sites that yielded the material used in this study ($N = 177$, but note that some localities have been amalgamated for visual purpose).

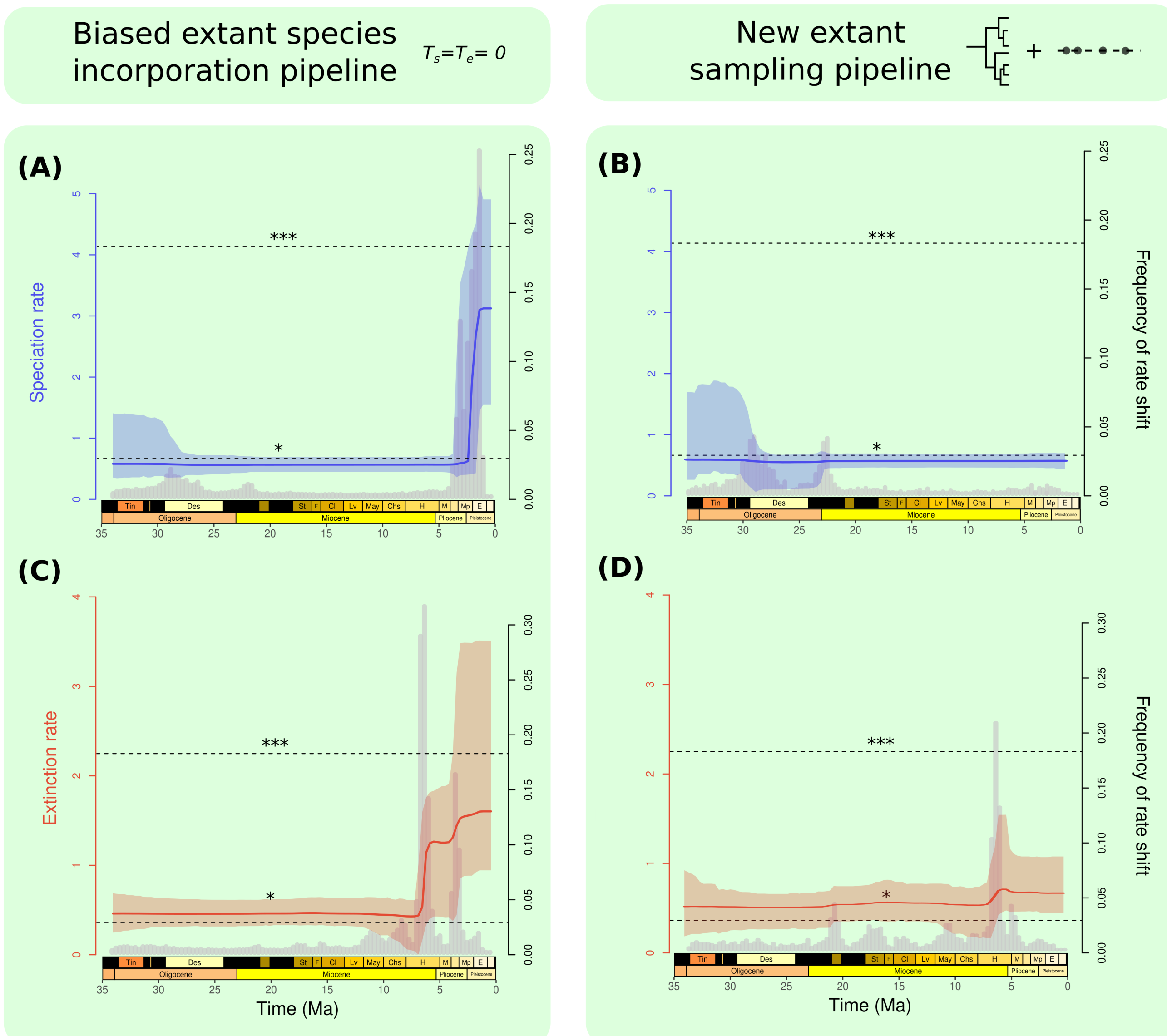

Figure S 2. Impact of the sampling scheme of extant species lacking fossil record on the estimation of speciation and extinction rates. Rates were estimated with PyRate using the rjMCMC algorithm, either from a dataset in which extant taxa were included by means of an occurrence with origination and extinction ages (T_e and T_e) set to 0 Ma (**A** and **C**) or with our newly-introduced pipeline relying on ages extracted from dated phylogenies (**B** and **D**). Full lines indicate median rate estimation (blue for speciation, red for extinction), and ribbons their associated 95% HPD interval. Grey histograms in the background depict the frequency of rate shift per 1-My time bins. The two horizontal lines labelled * and *** respectively indicate significant and strongly supported rate shift. Note that, in the two cases, these results were obtained prior to the incorporation of the taxonomic revision of the 'Mesopotamian' dinomyid fauna (Ituzaingó Fm.) made by Rasia (2026) (see Figure S12 for the subsequent effect of incorporating this revision on diversification rate estimation).

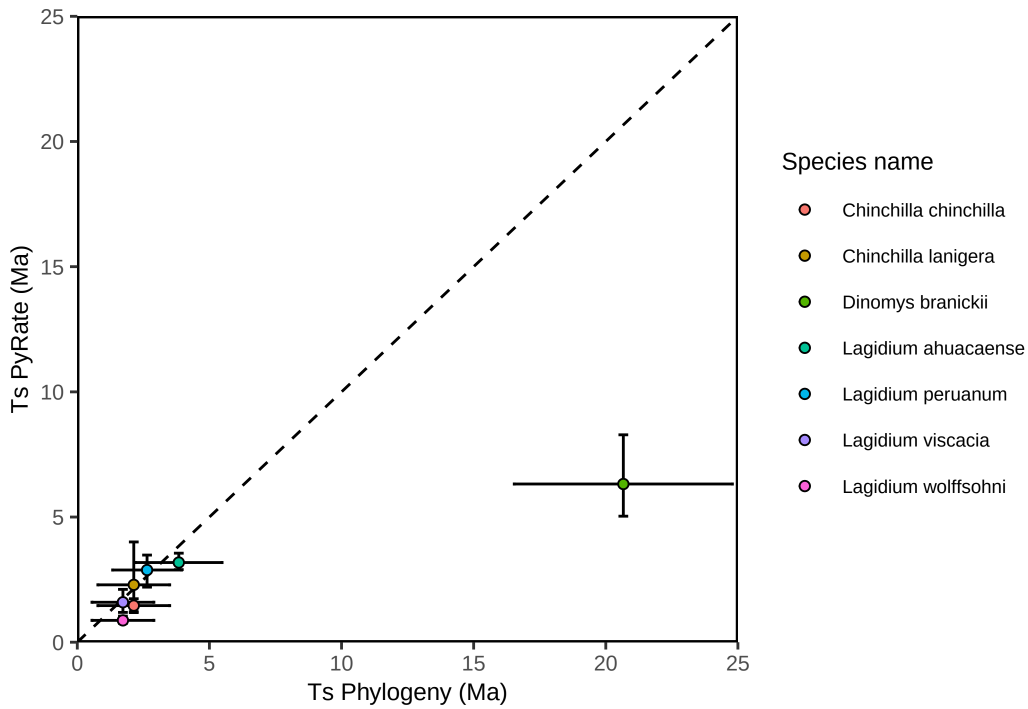

Figure S 3. Correspondance between divergence times leading to extant species estimated from our phylogeny ('Ts Phylogeny') and the times of species origination de novo estimated ('Ts PyRate') after implementing our sampling scheme for extant species absent from the fossil record (*Main text*, Figure 2). For each species, we set $\alpha = 0.95$, except for *D. branickii*, for which we set $\alpha = 0.999$. Dots show median estimates and error bars the associated 95% HPD. Dashed line shows the line with equation $y = x$.

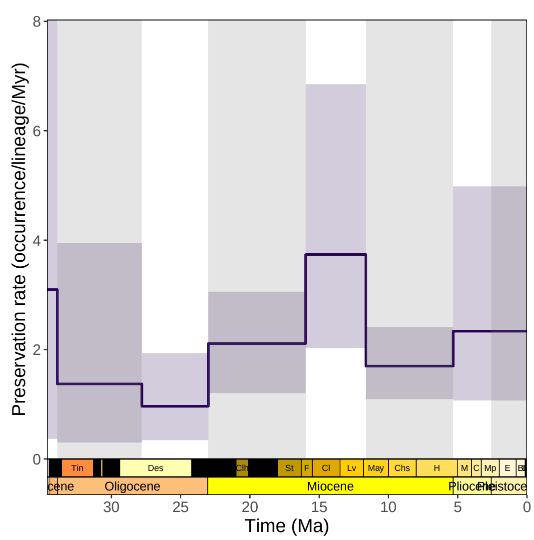

Figure S 4. Preservation rate through time in our species-level dataset estimated with PyRate. The solid line represents the mean posterior estimate and the shaded area the associated 95% HPD interval.

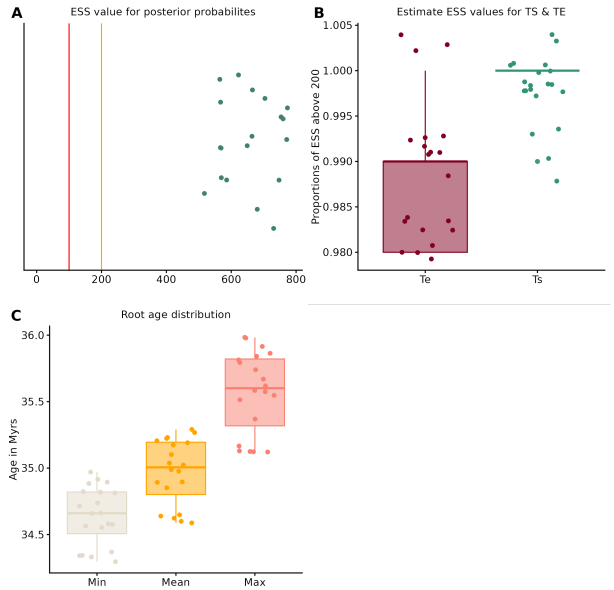

Figure S 5. Convergence of the PyRate runs (rjMCMC algorithm) for the analysis of the species-level temporal occurrence database. The plot (A) the ESS of the model posterior probability for all the 20 replicates we ran. ESS greater than 200 (first-scale indicator of a run that converged, $n = 20$ in that case) are displayed in green. The proportion of origination and extinction times (resp. T_S and T_E) that converged (i.e., with $ESS > 200$) are shown in (B), for each of our 20 replicates. The plot (C) shows the posterior estimates of the mean root age of our group, together with their lower and upper boundaries of the 95% HPD interval, across the 20 replicates.

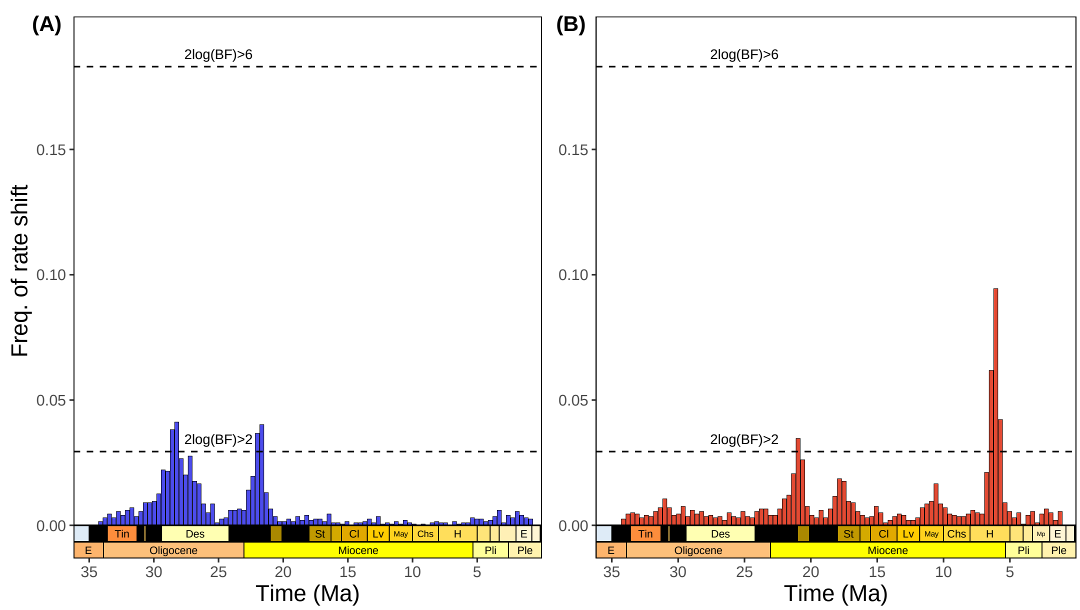

Figure S 6. Histogram showing the frequency of speciation (**A**) and extinction (**B**) rate shifts in our species-level chinchilloid dataset estimated by PyRate (rjMCMC algorithm). This quantity indicates, within 1My time bins, the support given to a model with rate shift compared to a model with constant rates. The two dashed lines respectively labelled $2log(BF) > 2$ and $2log(BF) > 6$ respectively indicate significant and strongly supported rate shift.

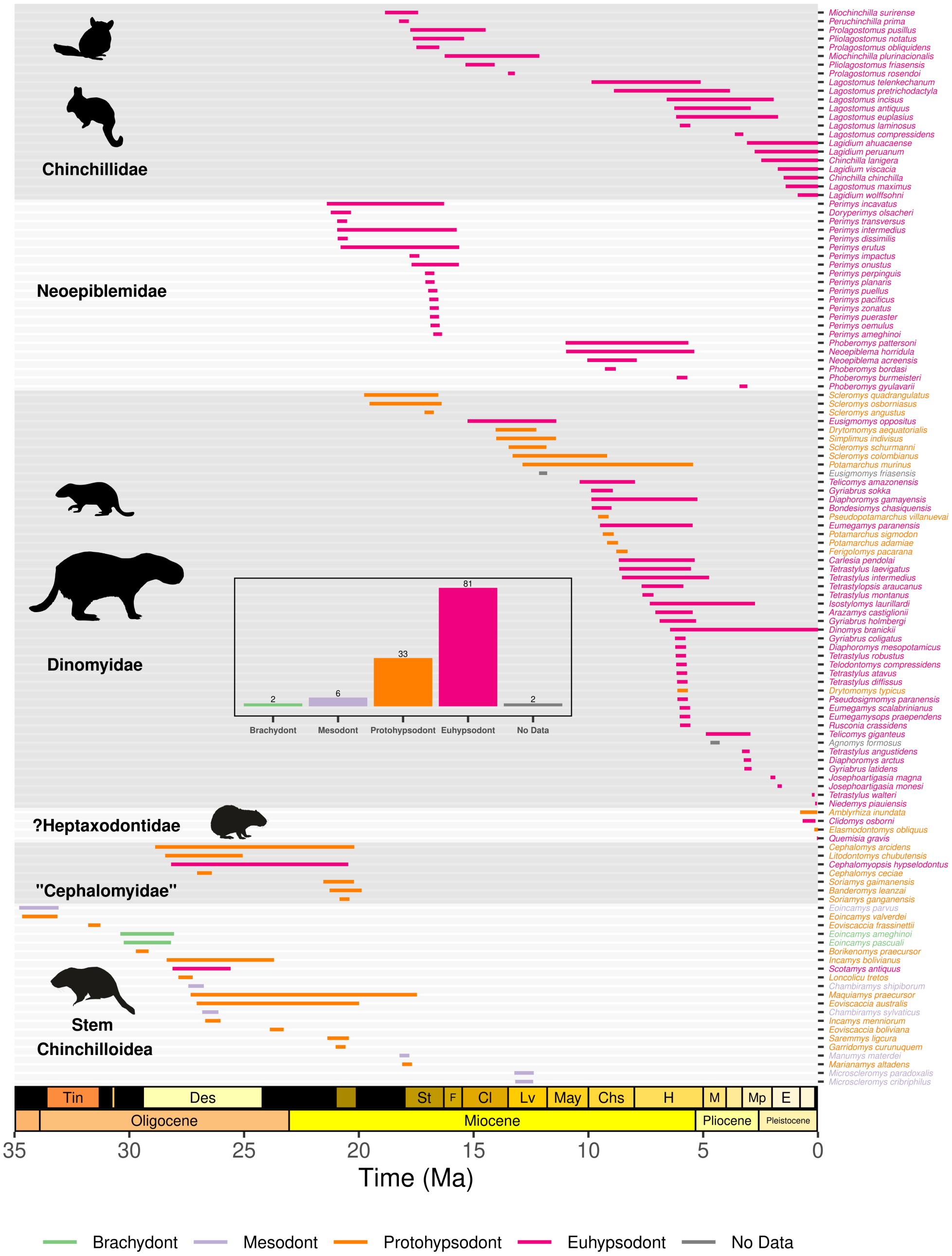

Figure S 7. Bayesian reconstruction of the longevities of the 124 chinchilloid species considered in our dataset. Each segment depicts the longevity of a given species, that is, the interval between species' speciation and extinction times averaged across PyRate runs that achieved convergence. Segments and taxonomic names were coloured according to the crown height category to which species were assigned. The histogram at the centre of the plot shows the number of species within each crown height class. The first four silhouettes (from top to bottom) are from Phylopic (credits available in Table S3). The last two silhouettes (*Elasmodontomys obliquus* and *Borikenomys praecursor*) were designed by L.M. for a previous work (Marivaux et al. 2020).

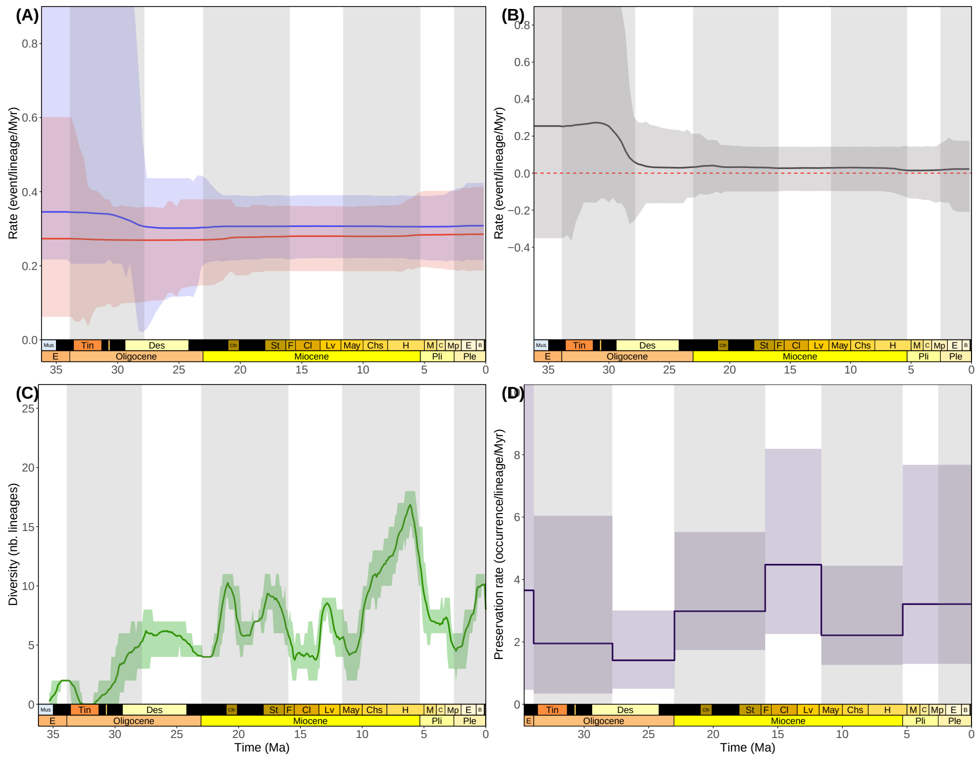

Figure S 8. Joint estimation of origination (blue) and extinction (red) rates (**A**), net diversification rate (**B**), diversity through time (**C**) and preservation rates (**D**) from our species-level chinchilloid database made my PyRate (rjMCMC algorithm) after removing singletons.

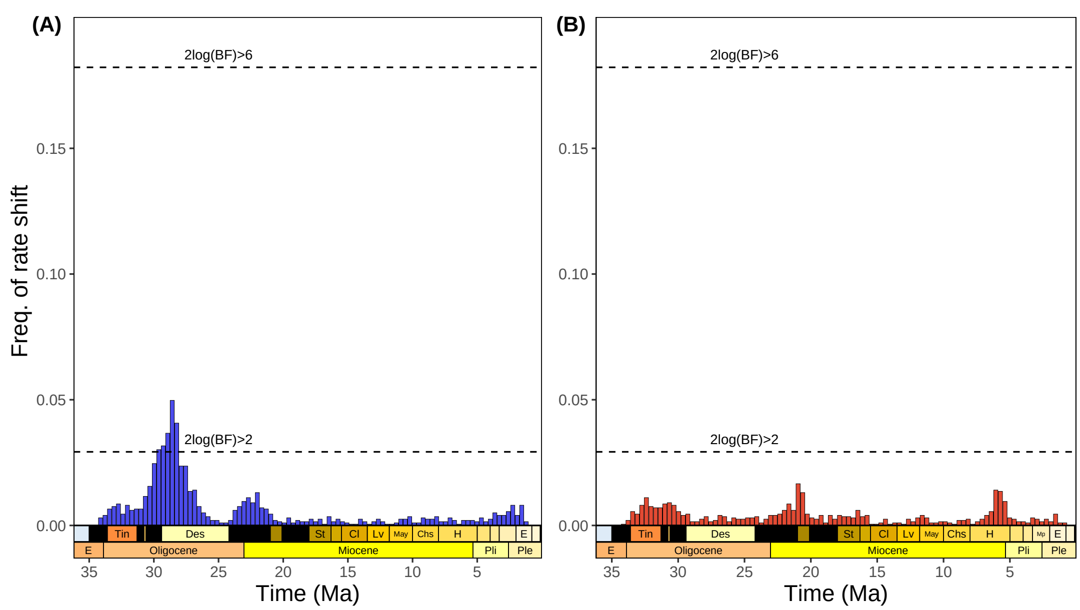

Figure S 9. Frequency of origination (**A**) and extinction (**B**) rates estimated by the rjMCMC analysis of our species-level database after removing singletons.

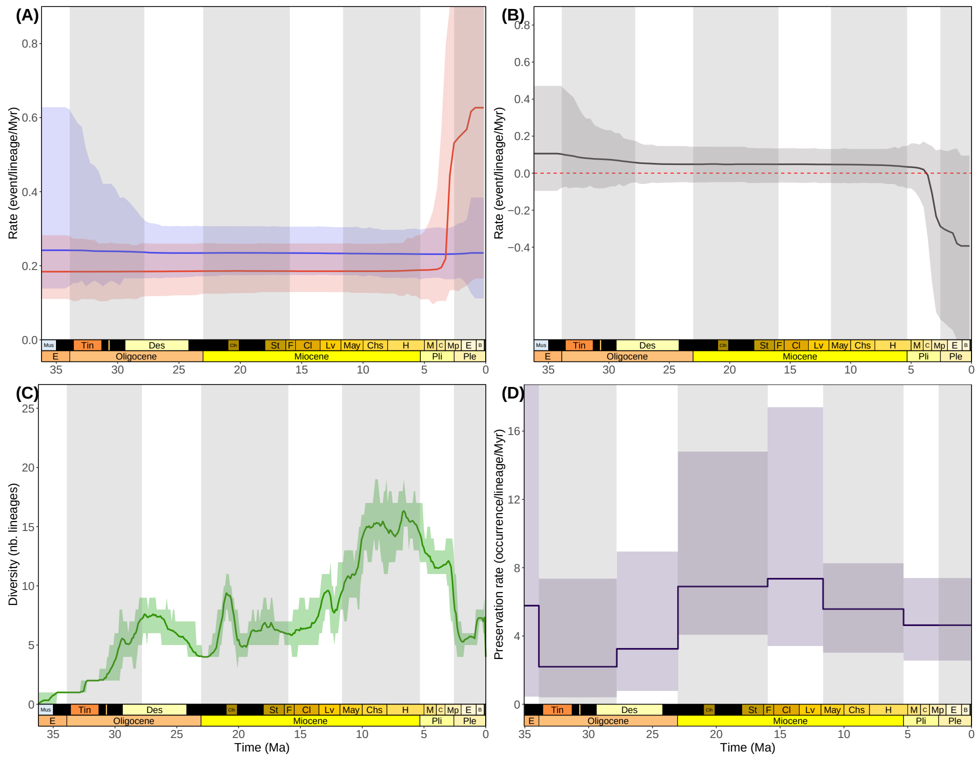

Figure S 10. Joint estimation of origination (blue) and extinction (red) rates (**A**), net diversification rate (**B**), diversity through time (**C**) and preservation rates (**D**) from the genus-level chinchilloid database made my PyRate (rjMCMC algorithm).

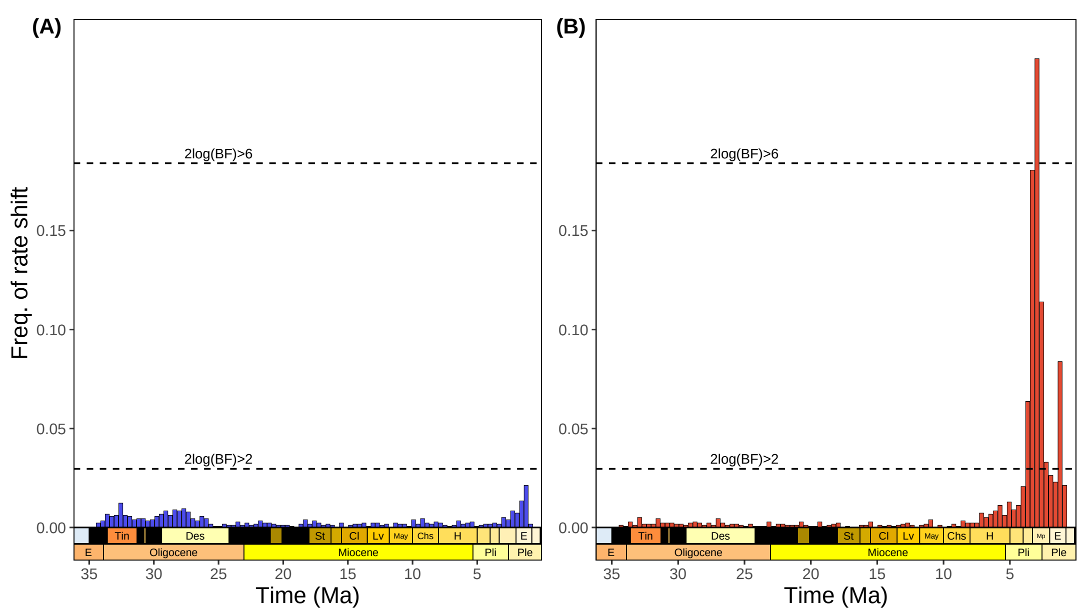

Figure S 11. Frequency of origination (**A**) and extinction (**B**) rates estimated by the rjMCMC analysis of our genus-level database.

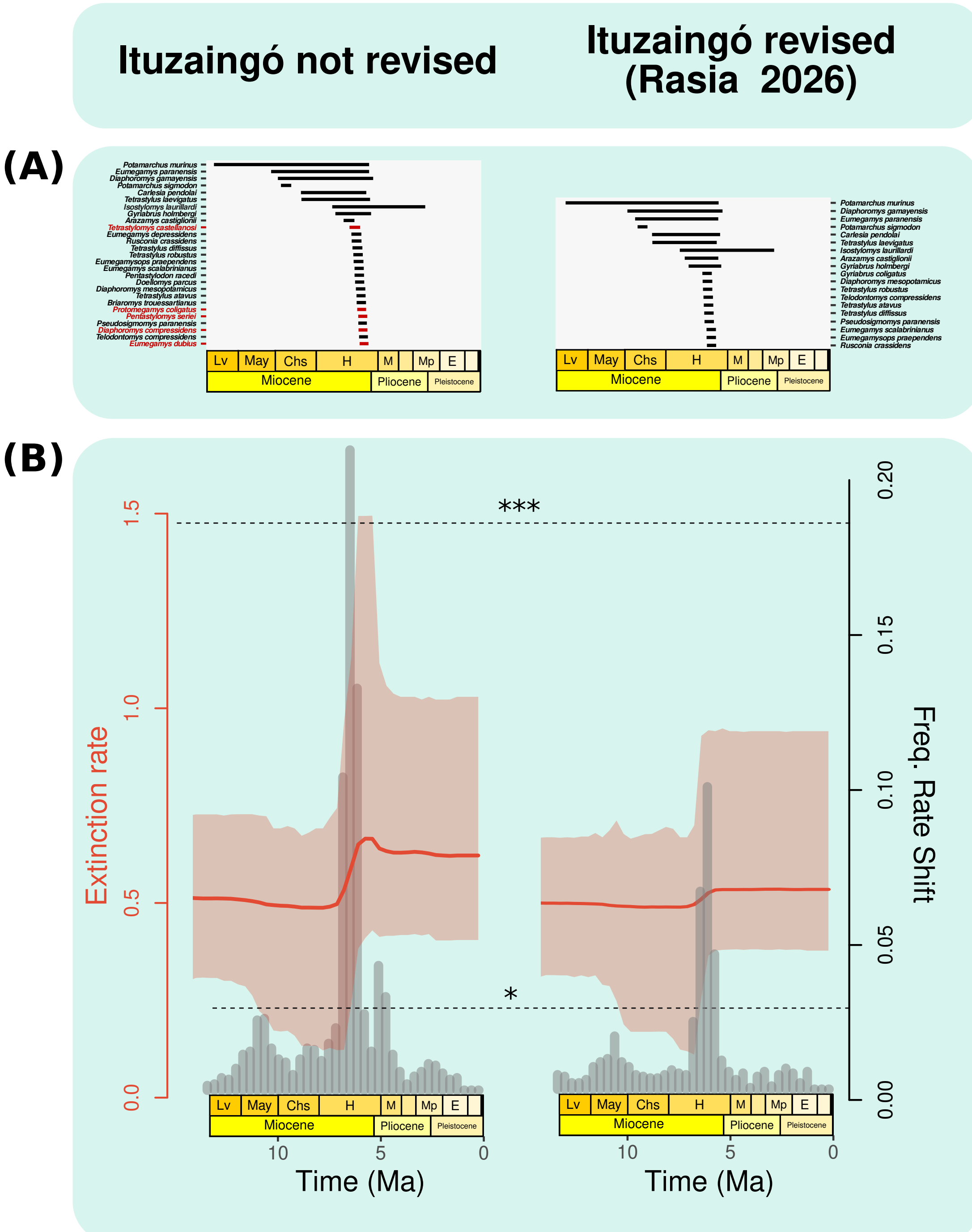

Figure S 12. Impact of occounting for the revision of the dinomyid fauna of the Ituzaingó formation (Entre Ríos province, Argentina) on the estimation of the temporal dynamics of chinchilloid extinction rate (Rasia 2026). (**A**) Lifespans of species making up the 'Mesopotamiense' fauna (Ituzaingó Fm.) before (*left*) and after (*right*) the taxonomic revision. Removed taxa were coloured in red (five taxa lost). (**B**) Chinchilloid extinction rates estimated from our temporal occurrence database before (*left*) and after (*right*) incorporating the aforementioned revision. The histogram in background shows the frequency of rate shift estimated by the rjMCMC model, that is, for a given time frame (by default set to 1My), how likely it is to consider that the reconstructed extinction rate experienced a shift. The dashed lines labelled (*) and (***) respectively indicate significant and strongly supported rate shift.

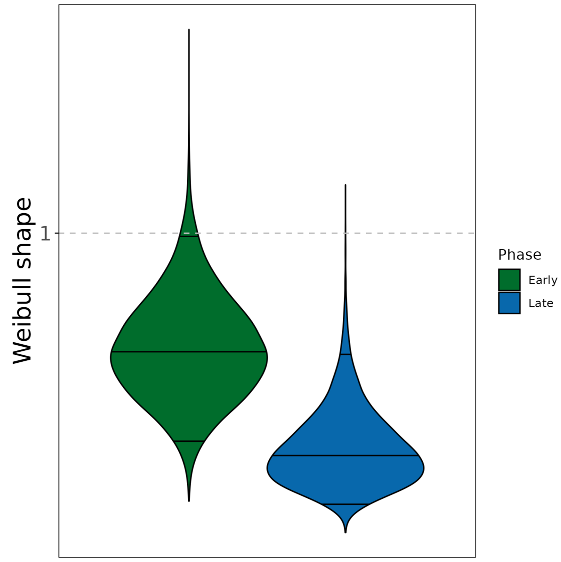

Figure S 13. Posterior estimate of the shape parameter of the Weibull distribution for the ADE analyses of the 'Early' (> 6 Ma) and 'Late' (< 6 Ma) occurrence datasets, each with constant extinction rate (Figure S6). 2.5%, 50% and 97.5% quantiles are indicated in the distribution of each parameter by black horizontal lines, respectively from bottom to top.

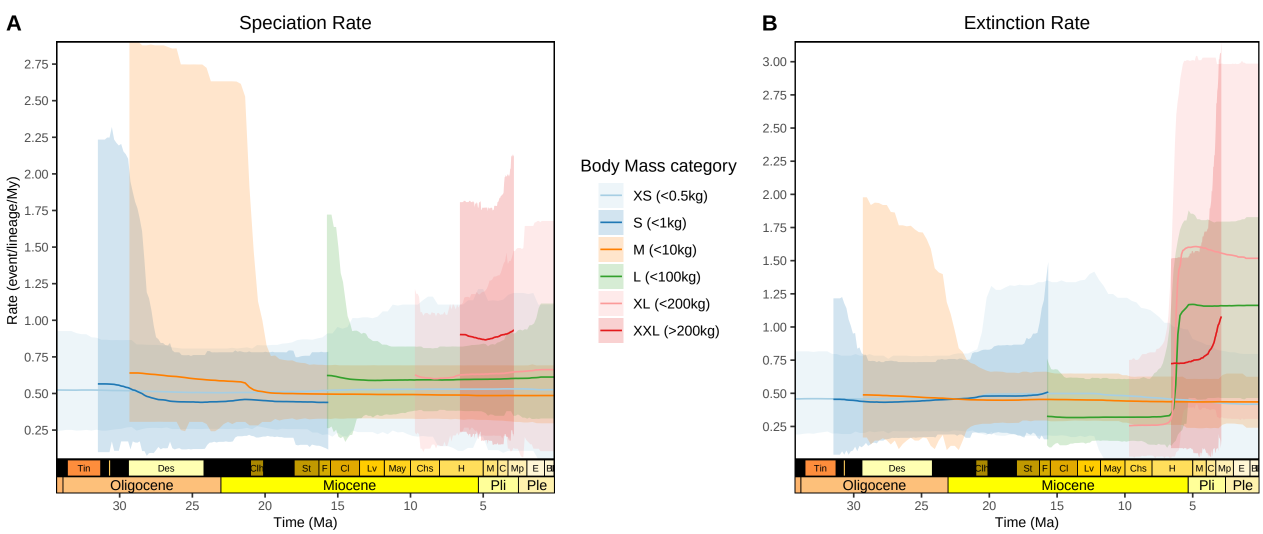

Figure S 14. Speciation (**A**) and extinction (**B**) rates through time per body mass class as inferred by the rjMCMC model.

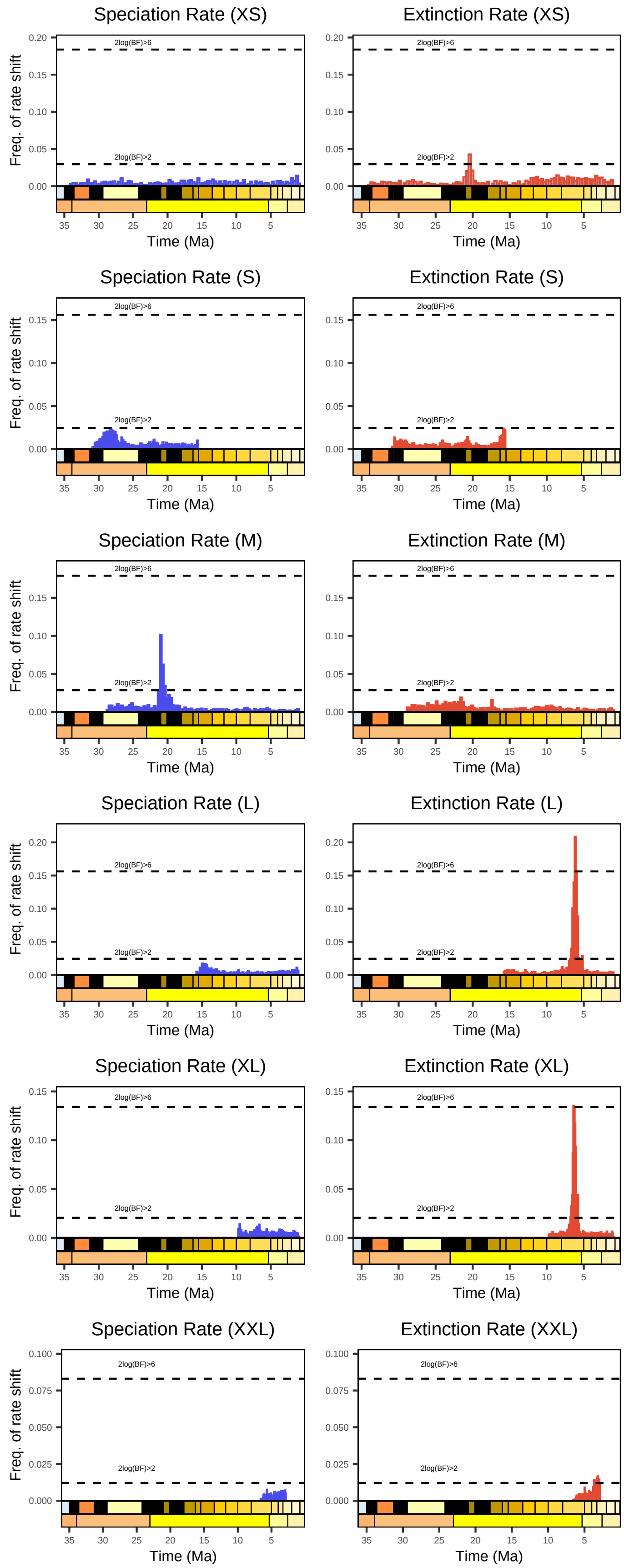

Figure S 15. Frequency of speciation and extinction rate shifts per body size class as estimated by the analysis of our species-level chinchilloid occurrence dataset with the rjMCMC model. Dashed horizontal lines indicate 2 * log(Bayes factor) of 2 and 6, respectively suggesting significant and strongly-supported rate shifts.

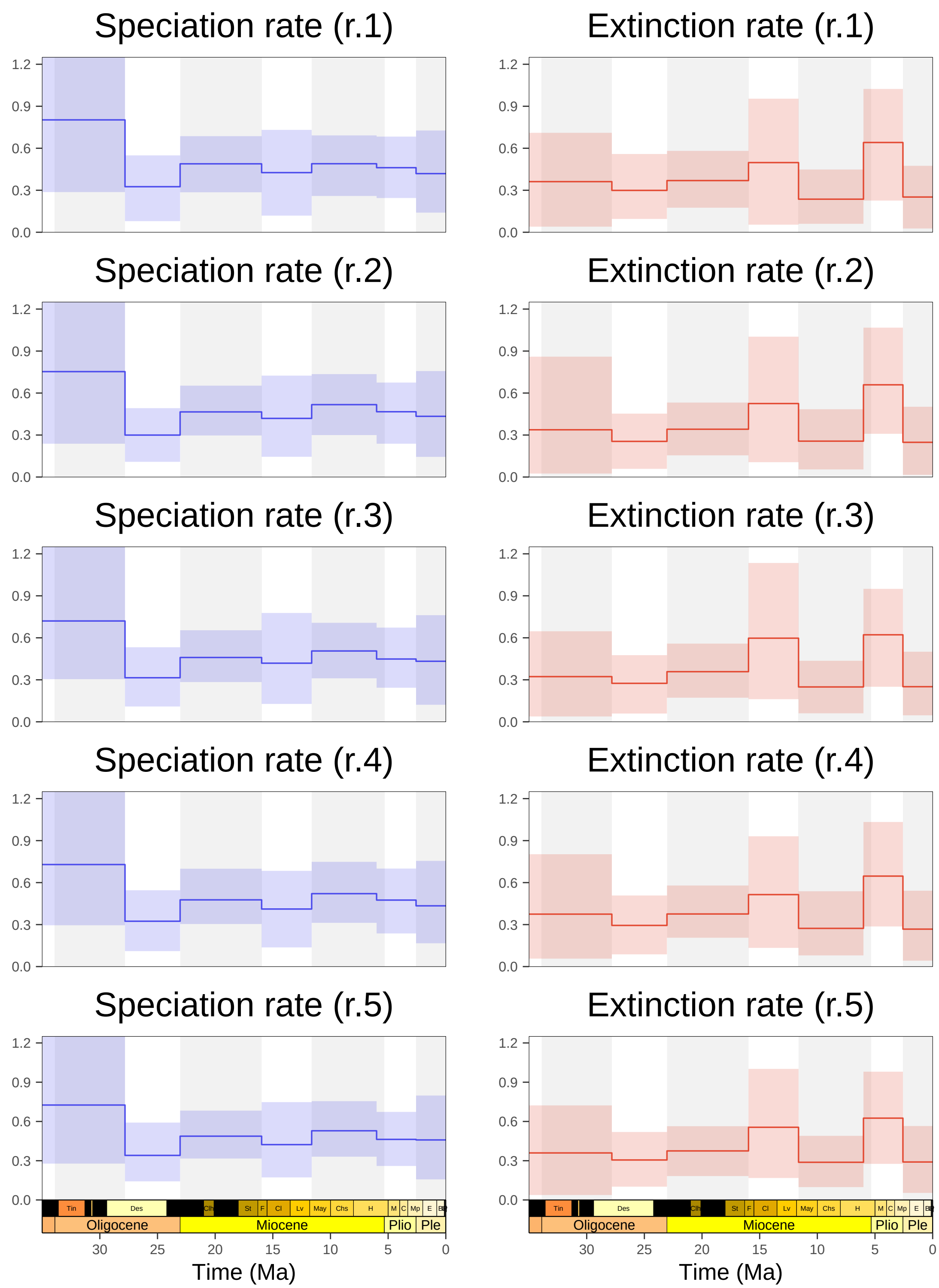

Figure S 16. Marginal chinchilloid speciation (blue) and extinction (red) rates through time as estimated by our BDNN analyses of the datasets with random trait (*i.e.*, body mass and/or hypsodonty index) imputation for taxa with missing trait reconstruction ($n = 11$). Caribbean taxa were excluded. Full lines show median rate estimates and ribbons their associated 95% Highest Posterior Density (HPD) interval, assessed acrossed the 10 independent BDNN runs randomising fossil ages under a uniform law made for each trait imputation. This panel only depicts the rates through time from replicate 1 to 5.

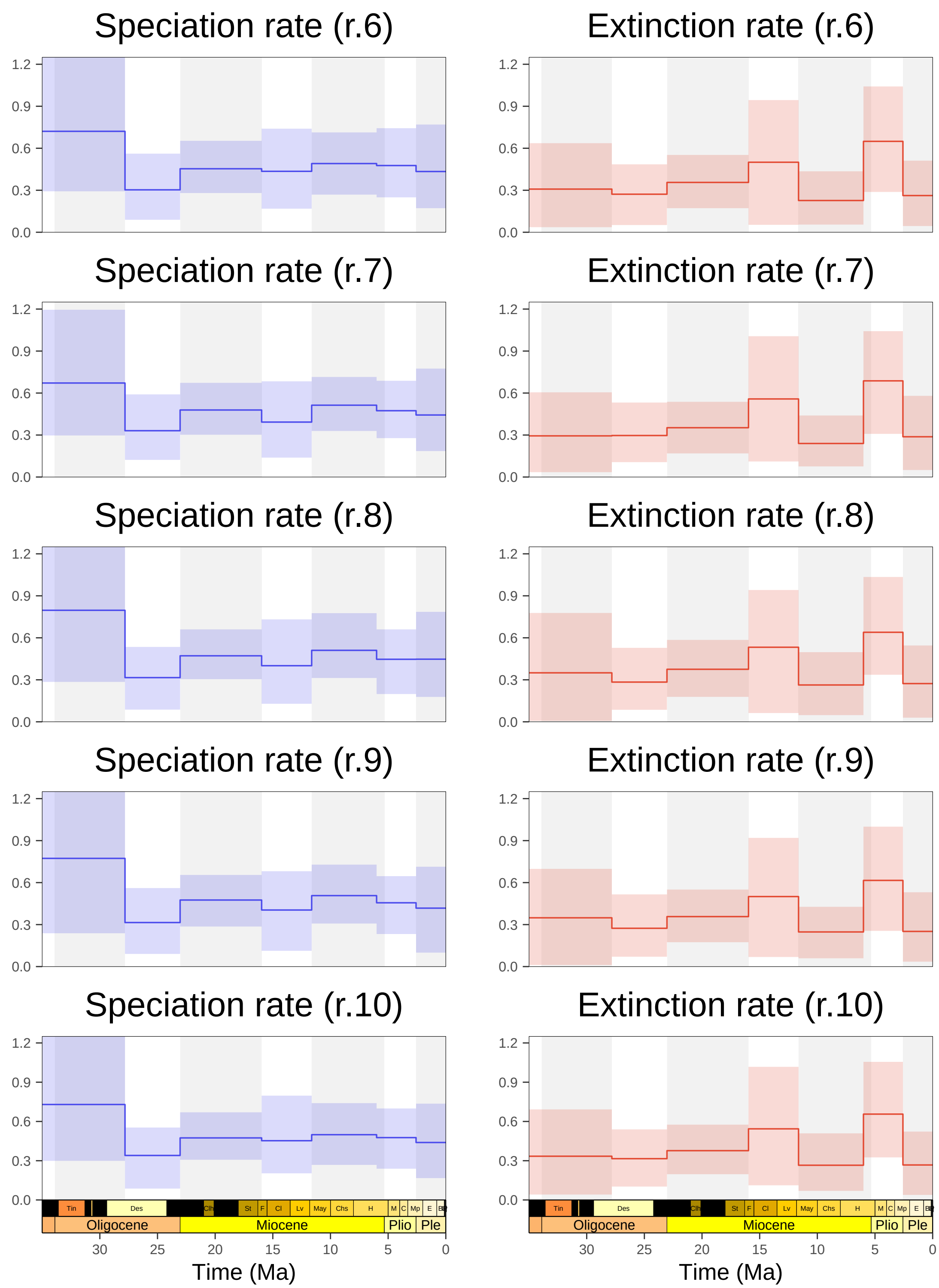

Figure S 17. Marginal chinchilloid speciation (blue) and extinction (red) rates through time as estimated by our BDNN analyses of the datasets with random trait (*i.e.*, body mass and/or hypsodonty index) imputation for taxa with missing trait reconstruction ($n = 11$). Caribbean taxa were excluded. Full lines show median rate estimates and ribbons their associated 95% Highest Posterior Density (HPD) interval, assessed acrossed the 10 independent BDNN runs randomising fossil ages under a uniform law made for each trait imputation. This panel only depicts the rates through time from replicate 1 to 5.

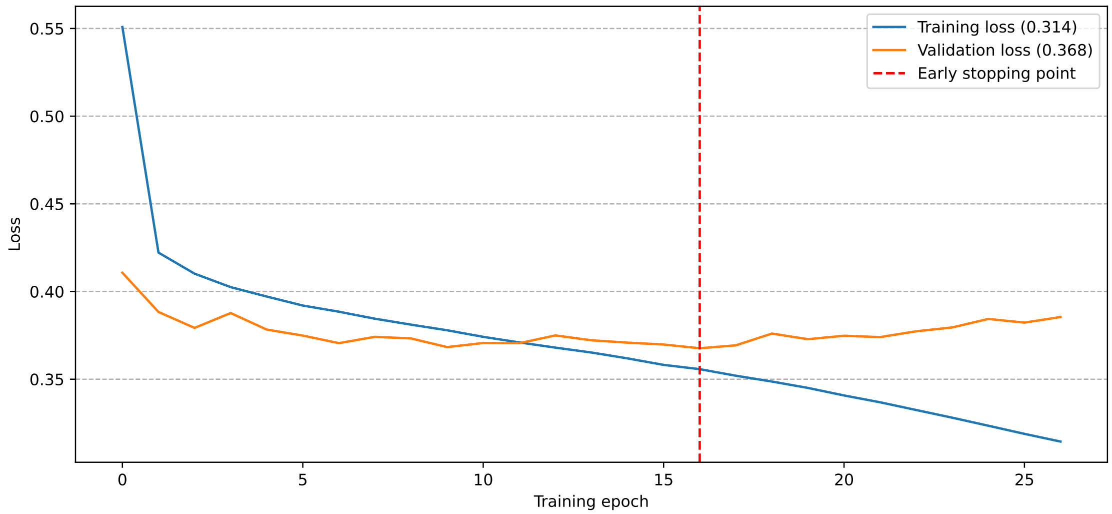

Figure S 18. Training and validation loss across training epochs for a DeepDive network comprising two LSTM units (64 and 32 nodes) connected to two dense layers (64 and 32 nodes) before the output layer. The red vertical line indicates the "early stopping point", *i.e.*, when the validation loss starts increasing, indicative of overfitting. It determines the optimal value of number of iterations, at which we derive the training and validation loss values depicted between parenthesis.

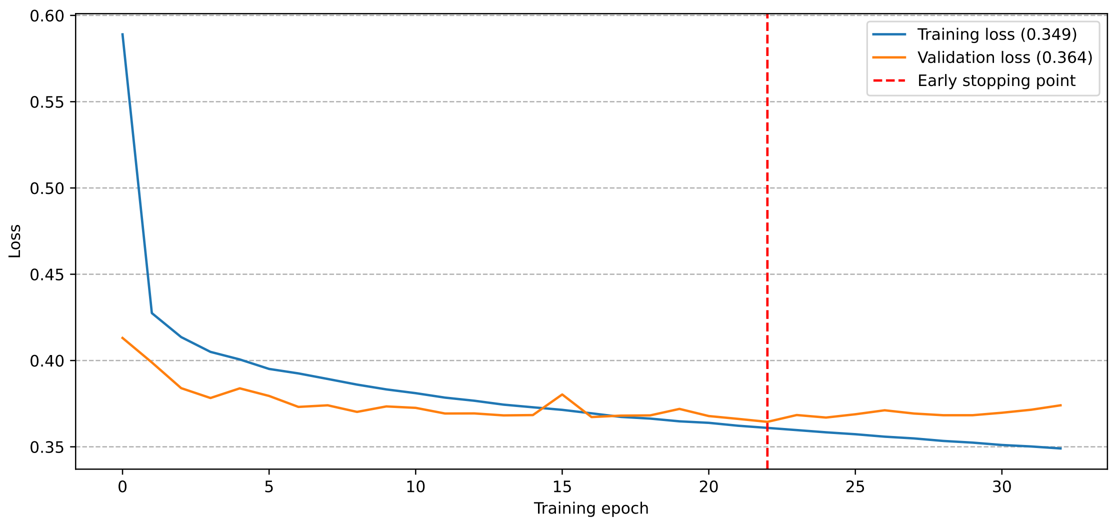

Figure S 19. Training and validation loss across training epochs for a DeepDive network comprising three LSTM units (128, 64 and 32 nodes) connected to two dense layers (64 and 32 nodes) before the output layer. The red vertical line indicates the "early stopping point", *i.e.*, when the validation loss starts increasing, indicative of overfitting. It determines the optimal value of number of iterations, at which we derive the training and validation loss values depicted between parenthesis.

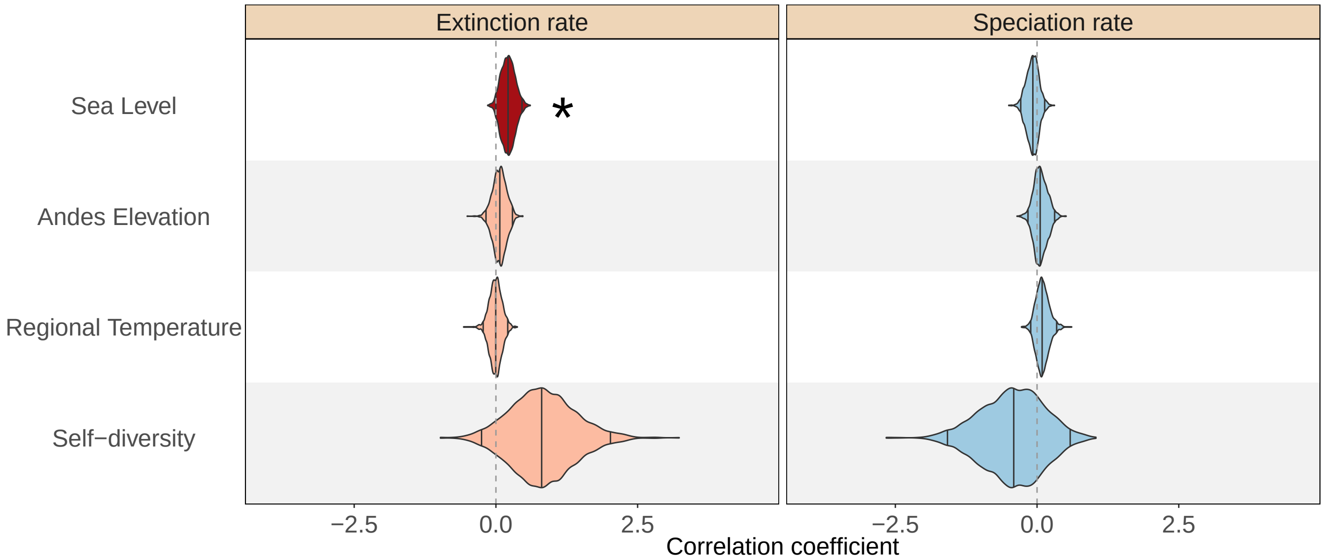

Figure S 20. Bayesian estimation of correlation coefficients of extinction (*left*) and speciation (*right*) rates with a selected set of environmental variable carried out by the MBD model (*see Methods in the main text*). A gamma prior was set to correlation parameters. Light colours (respectively blue for speciation and red for extinction) indicate non-significant correlations, and plain colours indicate significant correlations. A correlation was considered to be significant if 0 was not included in the parameter 95% HPD interval.

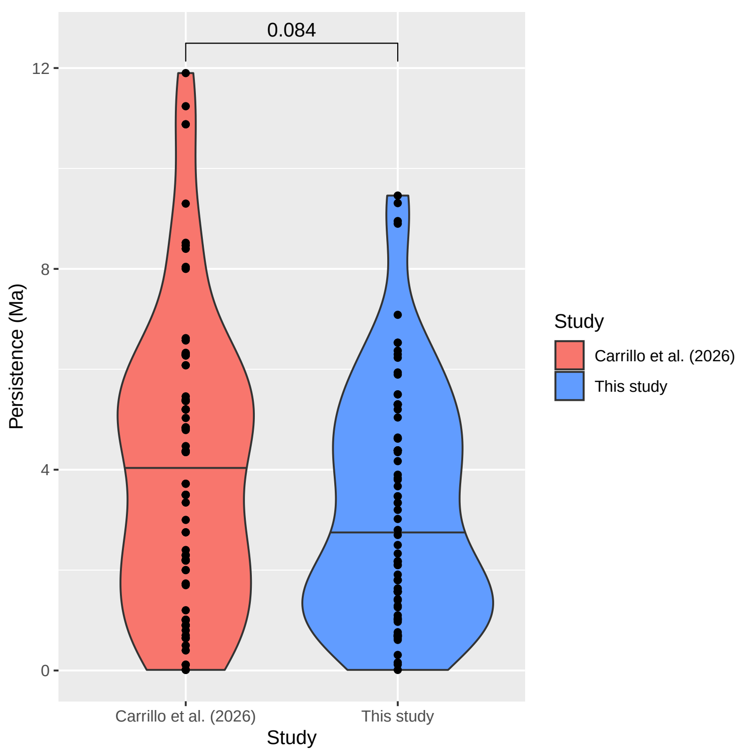

Figure S 21. Comparative persistence (*i.e.*, FAD minus LAD) of the 84 taxa in common between our database (in blue) and Carrillo et al. (2026). Mean persistences between the two studies were compared with a paired Wilcoxon test.

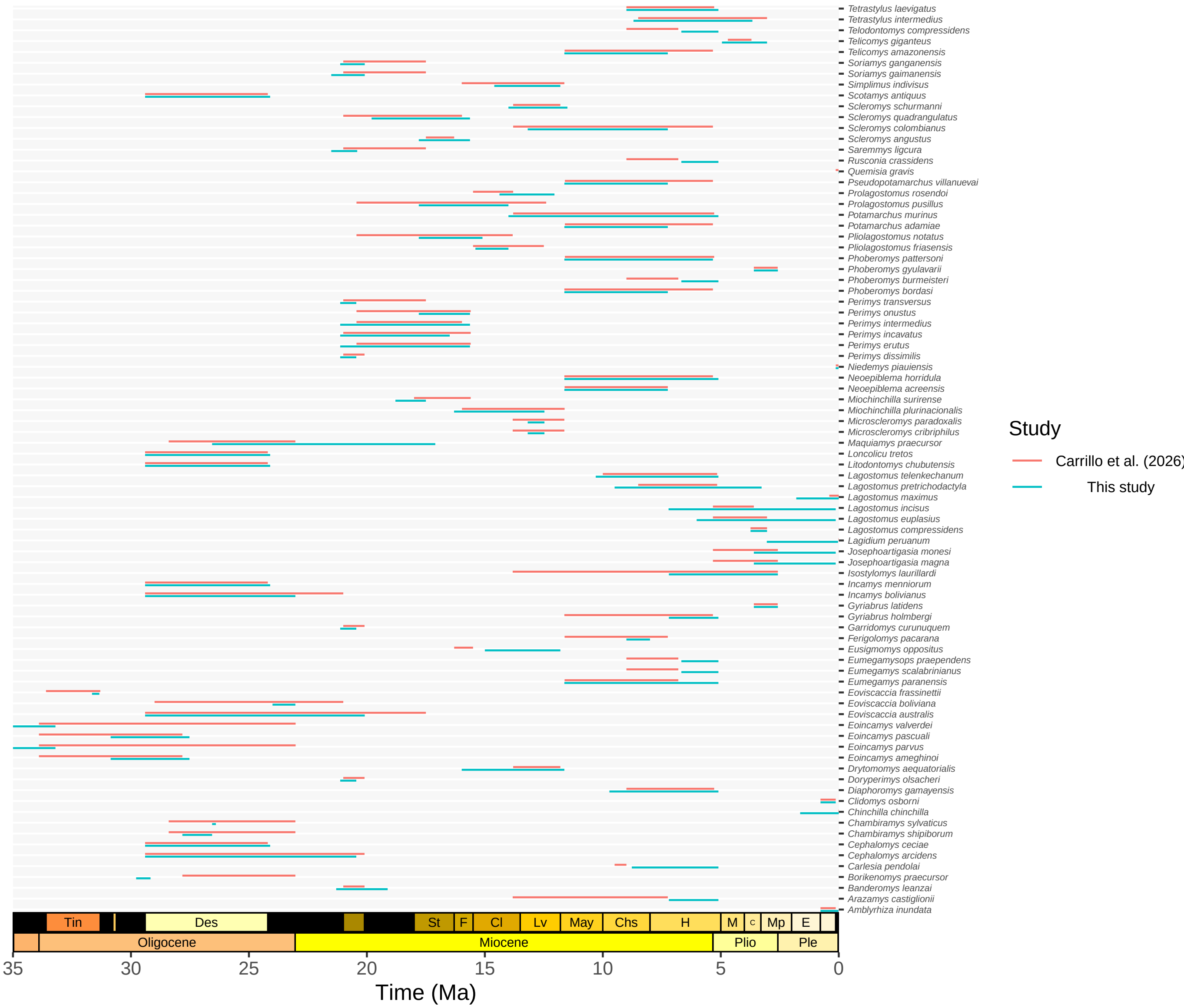

Figure S 22. Persistence in the fossil record of the 84 species shared between our database (in blue) and Carrillo et al. (2026) (in red). Persistence is assessed as the interval between the First and Last Appearance Data (resp. FAD and LAD) of each taxon.

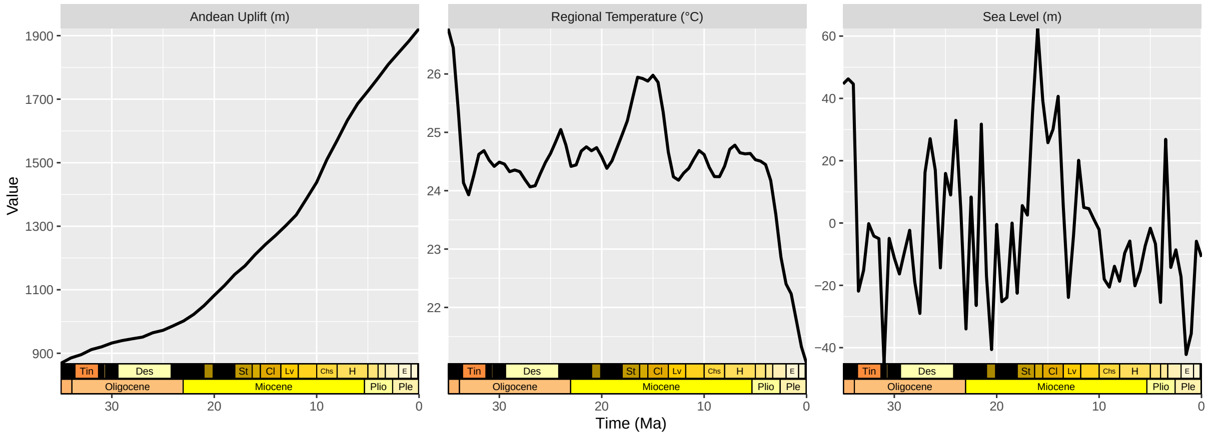

Figure S 23. Abiotic variables used in the environment-dependent diversification analyses carried out over this study. These have been temporally scaled to the same resolution (500ky-step). Andean uplift was from Boschman (2021), South American regional temperature was from Tardif et al. (2025) and sea-level changes were from Miller et al. (2020).

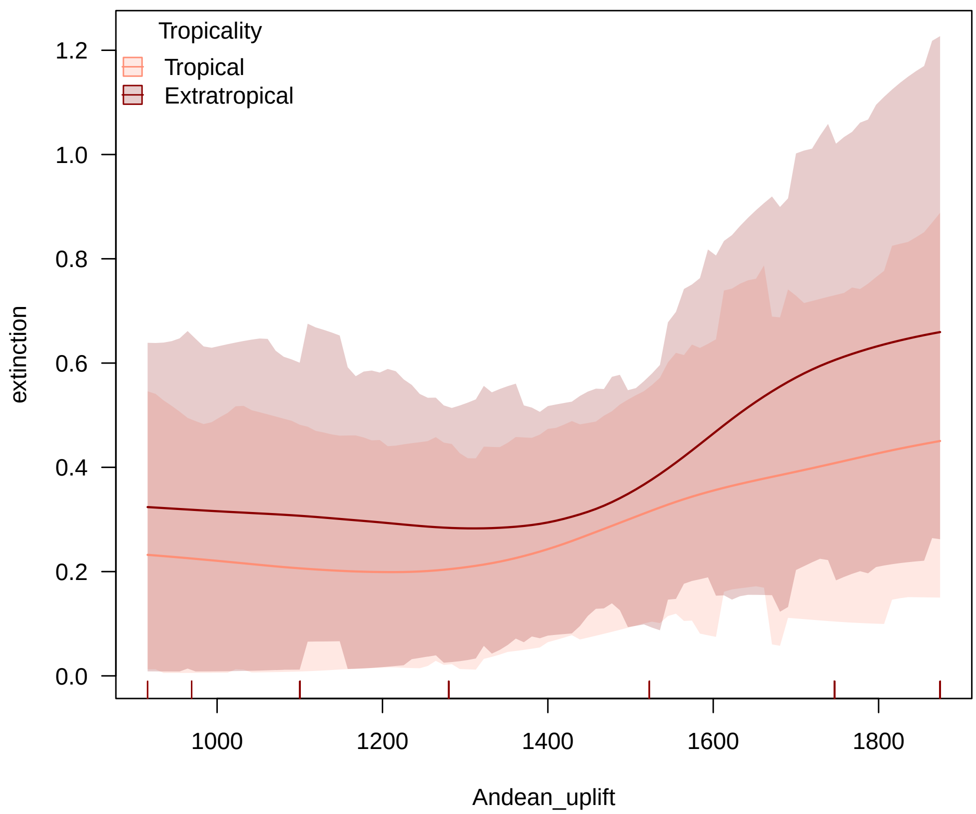

Figure S 24. Interaction between latitude and Andean uplift in shaping chinchilloid extinction rate variations across lineages, *i.e.*, how the relationship between the Andean uplift and extinction rate changes between tropical and extratropical species.

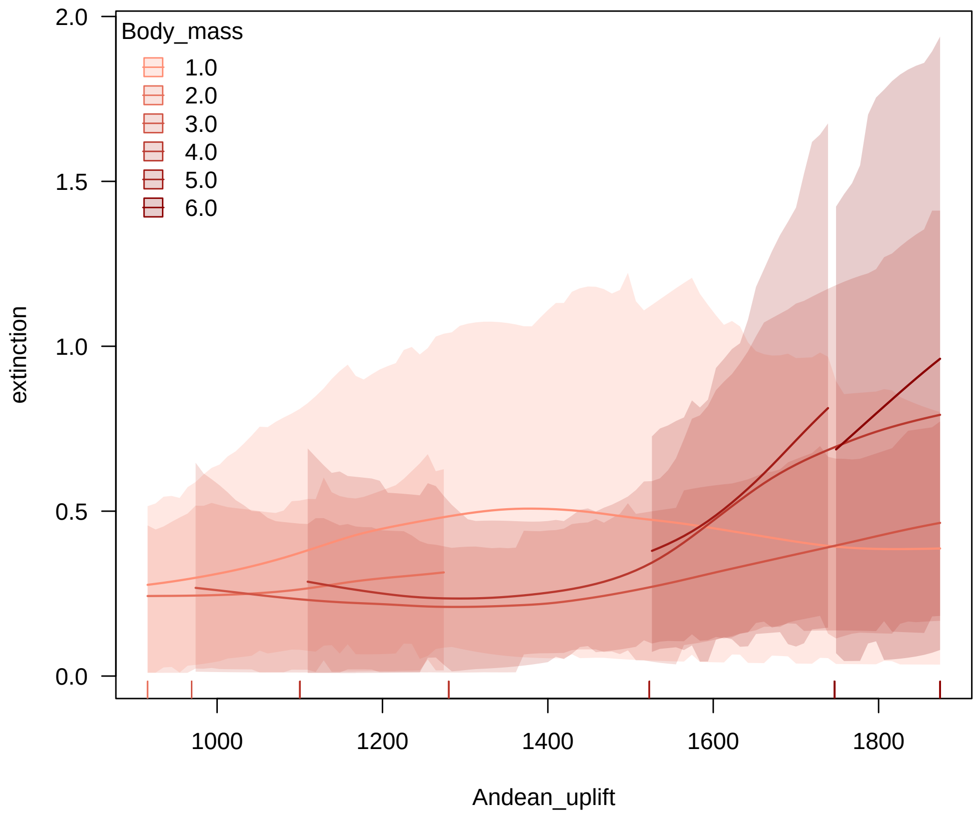

Figure S 25. Interaction between body mass and Andean uplift in shaping chinchilloid extinction rate variations across lineages, *i.e.*, how the relationship between the Andean uplift and extinction rate changes between body mass categories.

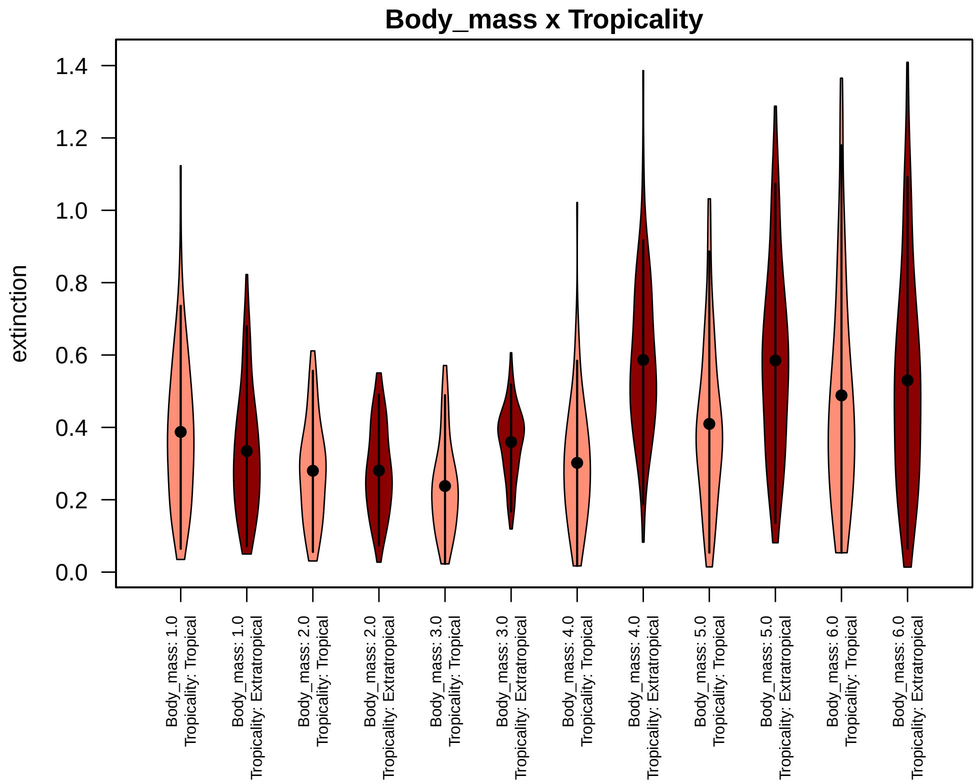

Figure S 26. Interaction between latitude and body mass in shaping chinchilloid extinction rate variations across lineages, *i.e.*, how sentitive to extinction body size classes are with respect to their tropical or extratropical origin.

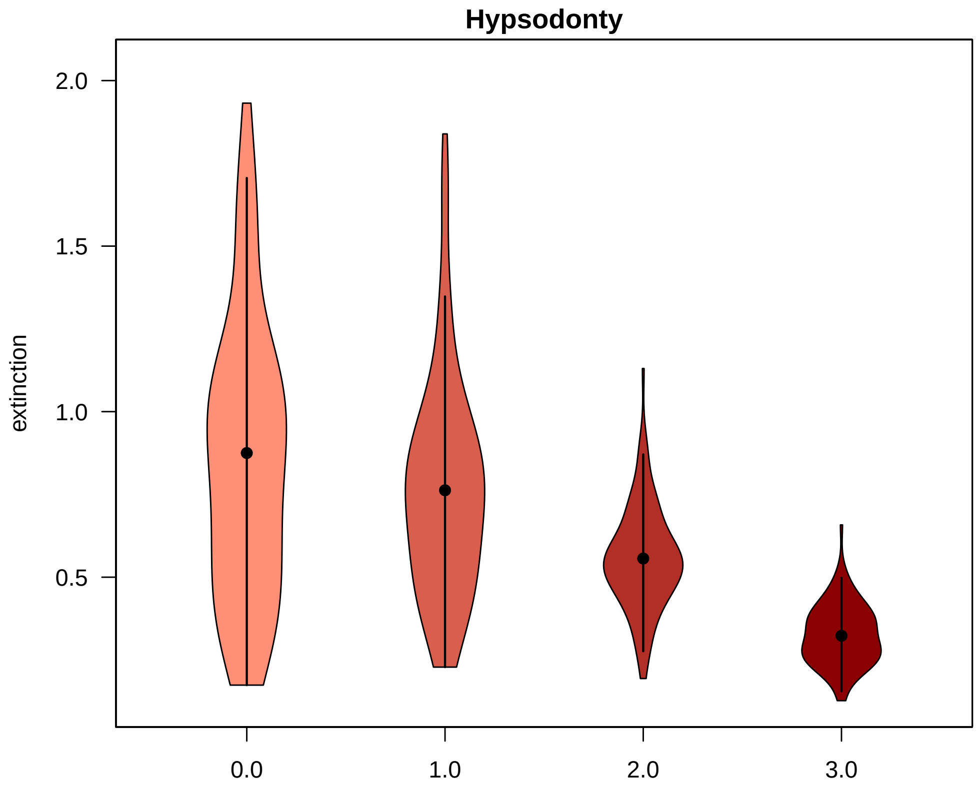

Figure S 27. How extinction rate varies across crown height categories (0: brachydont, 1: mesodont, 2: protohypsodont and 3: euhypsodont).

Supplementary Tables

Table S 1. Performance of the different network architectures tested in our DeepDive analyses. `LSTM Layers` and `Fully-connected` are respectively the number of Long Short Term Memory and Fully-connected units of the network (3: 128, 64 and 32 nodes; 2: 64 and 32 nodes; 1: 32 nodes). `Dropout` represents the Monte-Carlo dropout fraction, that is, the proportion of connections between neurons randomly toggled off between two learning epochs – one epoch being characterised by a complete pass through the entire training dataset —, preventing from overfitting. `Nb. training epochs` is the optimal number of training epochs. `Training loss` and `Validation loss` are the values of the Mean Squared Error (MSE) estimated between the 'true' diversity of each simulated dataset (resp. from the training or the validation set), and the predicted diversity made by the network after n rounds of optimisation, where n is the optimal number of training epochs. `Evaluation loss` represent the MSE between each true and its associated predicted diversity trajectory from the test datset, once the is trained (*i.e.*, after n rounds of optimisation, where n is the optimal number of training epochs). The optimal number of epochs is determined when the `Validation loss` starts increasing (see Figure S18 and Figure S19). Architectures resulting in the smallest evaluation loss values, *i.e.*, generating the most accurate predictions, are depicted in bold.

LSTM layers	Fully-connected	Dropout	Nb. training epochs	Training loss	Validation loss	Evaluation loss
3	**2**	**0.05**	**16**	**0.314**	**0.368**	**0.377**
3	1	0.05	14	0.324	0.367	0.38
2	**2**	**0.05**	**22**	**0.349**	**0.364**	**0.377**
2	1	0.05	19	0.354	0.366	0.38
1	1	0.05	41	0.373	0.37	0.381

Table S 2. Table representing the lower and upper bounds of the 95% HPD (resp. 'q_2.5%' and 'q_97.5%') of the distribution of the divergence ages leading to all the extant chinchilloid species absent from the fossil record – *i.e.*, all species but *Lagostomus maximus*. Ages are expressed in Ma.

	Chinchilla chinchilla	*Chinchilla lanigera*	*Dinomys branickii*	*Lagidium ahuacaense*	*Lagidium peruanum*	*Lagidium viscacia*	*Lagidium wolffsohni*
q_2.5%	0.78	0.78	16.54	2.20	1.34	0.56	0.56
q_97.5%	3.50	3.50	24.79	5.48	3.94	2.90	2.90

Table S 3. References of the silhouettes used in Figure 1, from Phylopic.

Silhouette	**Author**	**License**
Chinchilla chinchilla	Margot Michaud	CC0 1.0
Lagidium peruanum	Josue Hermoza	CC BY-SA 3.0
Dinomys branickii	Juan C. Cepeda Duque	CC BY 4.0
Josephoartigasia monesi	Nobu Tamura (body), Gustavo Lecuona (head)	CC BY-SA 3.0

Table S 4. Coefficient of speciation rate variation across the 10 BDNN replicates with random trait (*i.e.*, body mass and/or hypsodonty index) imputation for taxa with missing trait reconstruction ($n = 11$), excluding caribbean taxa. Simulated values (`cv_simulated`) were obtained from the analysis of 100 simulated datasets matching the empirical datasets in terms of number of traits, clade age and (approximately) number of extant species, generated under a constant-rate birth-death model. Empirical values (`cv_empirical`) were obtained from the analysis of the empirical dataset.

cv_empirical	cv_expected	Replicate
0.25	0.2	1
0.26	0.25	2
0.27	0.23	3
0.24	0.26	4
0.23	0.27	5
0.26	0.2	6
0.23	0.25	7
0.27	0.22	8
0.28	0.21	9
0.24	0.23	10
0.25	**0.23**	**Total**

Table S 5. Coefficient of extinction rate variation across the 10 BDNN replicates with random trait (*i.e.*, body mass and/or hypsodonty index) imputation for taxa with missing trait reconstruction ($n = 11$), excluding caribbean taxa.

cv_empirical	cv_expected	Replicate
0.52	0.31	1
0.52	0.34	2
0.49	0.37	3
0.48	0.37	4
0.47	0.35	5
0.53	0.38	6
0.5	0.39	7
0.5	0.38	8
0.53	0.28	9
0.49	0.34	10
0.51	**0.35**	**Total**

Table S 6. Predictor importance on extinction rate across the 10 BDNN replicates with random trait (*i.e.*, body mass and/or hypsodonty index) imputation for taxa with missing trait reconstruction (n = 11) after excluding caribbean taxa. We highlighted in bold the three factors that contribute the most to variations in extinction rates across species.

Predictor	R. 1	R. 2	R. 3	R. 4	R. 5	R. 6	R. 7	R. 8	R. 9	R. 10	SumRank	Rank
Andean_uplift	1	1	1	1	1	1	1	1	1	1	10	**1**
Body_mass	2	2	2	2	2	2	2	2	2	2	20	**2**
Tropicality	4	4	3	3	4	3	3	4	4	3	35	**3**
Hypsodonty	3	3	5	4	3	4	4	3	3	4	36	4
Self_diversity	6	5	4	4	6	6	6	5	6	5	53	5
Temperature	5	6	6	5	5	5	5	6	5	6	54	6
Sea_level	7	7	7	7	7	7	7	7	7	7	70	7
time	8	8	8	8	8	8	8	8	8	8	80	8

Table S 7. Magnitude of the change in extinction rate (*i.e.*, `Fold Change`) between the two extreme body size classes showing the most pronounced extinction rate variation (S and XL) across the 10 trait-imputed replicates without Caribbean taxa. `Upper CI` and `Lower CI` respectively show the uppper and lower boundaries of the associated 95% HPD interval. The 'Total' row shows the averaged values across the 10 replicates.

Replicate	Fold Change	Upper CI	Lower CI
1	1.74	7.62	-2.95
2	1.86	7.11	-2.47
3	2.41	8.65	-1.96
4	2.06	8.05	-2.33
5	1.85	7.17	-2.45
6	1.65	6.89	-3.04
7	1.66	6.09	-3
8	1.89	8.17	-2.9
9	2.09	9.25	-2.07
10	1.89	6.64	-2.62
Total	**1.91**	**7.56**	**-2.58**

Table S 8. Magnitude of the change in extinction rate (*i.e.*, fold change) between tropical and extratropical taxa across the 10 trait-imputated replicates without Caribbean taxa. `Upper CI` and `Lower CI` respectively show the uppper and lower boundaries of the associated 95% HPD interval. The 'Total' row shows the averaged values across the 10 replicates.

Replicate	Fold Change	Upper CI	Lower CI
1	1.67	5.96	-1.52
2	1.52	5.09	-1.93
3	1.49	4.05	-1.94
4	1.41	3.98	-1.72
5	1.5	3.93	-1.67
6	1.55	5.14	-2.01
7	1.49	4.63	-1.71
8	1.64	4.71	-1.78
9	1.53	4.52	-1.95
10	1.46	4.39	-1.71
Total	**1.52**	**4.64**	**-1.79**

Table S 9. Posterior estimates of the multivariate birth–death (MBD) correlation parameters with the extinction rate (G_μ, median), shown along with their 95% Highest Posterior Density (HPD) interval, for each of the four environmental variable tested. The only significant correlation, *i.e.*, of which the 95% HPD of the associated coefficient does not significantly overlap with 0, is shown in bold.

Covar	G_mu	G_mu_95HPD
Self-diversity	0.81	[-0.25; 2.02]
Regional Temperature	0	[-0.22; 0.21]
Andes Elevation	0.07	[-0.18; 0.29]
Sea Level	**0.21**	**[0; 0.46]**

Table S 10. Posterior estimates of the multivariate birth–death (MBD) correlation parameters with the speciation rate (G_μ, median), shown along with their 95% Highest Posterior Density (HPD) interval, for each of the four environmental variable tested. As all HPD intervals overlap with zero, no ignificant correlation was highlighted by this analysis.

Covar	G_lamb	G_lamb_95HPD
Self-diversity	-0.41	[-1.58; 0.58]
Regional Temperature	0.09	[-0.11; 0.35]
Andes Elevation	0.06	[-0.16; 0.31]
Sea Level	-0.07	[-0.29; 0.13]

Table S 11. Median estimate and associated 95% HPD bounds of the shape parameter of the Weibull shape parameter (Φ) of the Age-Dependent Extinction analysis of our species-level chinchilloid temporal occurrence dataset. Since the ADE model should be run on periods of constant extinction rate through time, `Species_Early` and `Species_After` respectively designate the species before and after the shift in extinction rate occurring *ca.* 6 Ma.

Phase	Median_estimate	HPD95
Species Early	0.66	[0.4; 0.99]
Species Late	0.36	[0.21; 0.65]

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